CrimeScope

India Crime Analytics — Project Report

MySQL → pandas → Power Query → DAX → Power BI

12	35	806	29	23.9M
YEARS	STATES/UTS	DISTRICTS	CATEGORIES	TOTAL CASES

1 Project Overview

CrimeScope is an end-to-end crime analytics project built on 12 years (2001–2012) of India's National Crime Records Bureau (NCRB) IPC crime data. It demonstrates the complete analytics lifecycle used in real data-analytics and consulting work: sourcing data from a live database, validating and cleaning it programmatically, reshaping it for analysis, modeling it properly, and presenting findings through a designed, decision-ready dashboard.

> **Why this project** — Rather than starting from a pre-cleaned CSV, this project intentionally works from a raw MySQL source so it demonstrates the full pipeline a working analyst uses day to day, including the data-quality problems that real datasets actually contain.

The Pipeline

MySQL (source data) → **Python / pandas (validation)** → **Power Query (M) (ETL / reshape)** → **Star Schema (data model)** → **DAX (measures)** → **Power BI (dashboard)**

Dataset at a Glance

Attribute	Value
Source	MySQL database (schema: crime_analysis, table: crime_data)
Rows (raw, wide format)	8,597
Geographic coverage	35 States / Union Territories, 806 Districts
Time coverage	2001 – 2012 (12 years)
Crime category columns	29 (originally wide format — one column per crime type)
Missing values	0
Duplicate rows	0

Objectives

- Build a reusable, end-to-end pipeline from raw relational data to an interactive BI dashboard.
- Practice and demonstrate SQL sourcing, pandas validation, Power Query ETL, DAX, and dashboard/UX design in a single project.
- Surface a genuine, defensible data-driven insight — not just charts.
- Apply professional dashboard design standards: consistent theme, typography, KPI structure, multi-page navigation.

2 How This Was Built

This section documents the actual build process, in the order it happened, so the reasoning behind each decision is traceable.

Step 1 — Connect to the source

Power BI Desktop was connected directly to the MySQL database using the native connector, rather than working from an exported file. This kept the pipeline refreshable — if the underlying table changes, the whole model can be refreshed end-to-end.

```
Source = MySQL.Database("localhost:3306", "crime_analysis", [ReturnSingleDatabase=true]),
crime_analysis_crime_data = Source{[Schema="crime_analysis",Item="crime_data"]}[Data]
```

Step 2 — Validate before trusting (Python / pandas)

Before any modeling, the raw table was profiled independently in Python — shape, dtypes, null counts, duplicate rows, and category-level sums. This step caught a serious problem early: summing the 29 raw crime columns produced **28,037,178** total cases, while the dataset's own pre-computed `TOTAL_IPC_CRIMES` column summed to only **23,940,654** — a gap of over 4 million cases.

> **Root cause found via pandas aggregation:** six columns were sub-categories double-counted inside their own parent category — `CUSTODIAL_RAPE` and `OTHER_RAPE` inside `RAPE`; `KIDNAPPING_AND_ABDUCTION_OF_WOMEN_AND_GIRLS` and `_OF_OTHERS` inside `KIDNAPPING_AND_ABDUCTION`; and `AUTO_THEFT` / `OTHER_THEFT` inside `THEFT`. Excluding these six columns brought the reconciled total to exactly 23,940,654 — an exact match, confirming the fix rather than approximating it.

Step 3 — Reshape in Power Query

The raw table is wide (one column per crime type), which is unusable for category-level slicing or charting. It was unpivoted into a long fact table — one row per State–District–Year–Category — then cleaned and categorized:

- **Trim** — removed stray whitespace from `STATE_UT` / `DISTRICT` text fields.
- **Unpivot** — 29 wide columns → two columns, `CrimeCategory` and `Cases`.
- **Filter** — removed the `TOTAL_IPC_CRIMES` row and the six double-counted sub-category rows identified in Step 2.
- **Custom column** — added `CrimeGroups`, bucketing 23 categories into Violent Crime, Property Crime, Crime Against Women, Kidnapping/Abduction, and Other — turning a 29-row category list into a 5-group analytical story.

Step 4 — Build a star schema

Rather than leaving everything in one flat table, the model was split into a fact table and two dimension tables, related on `YEAR` and `STATE_UT` — the standard pattern for scalable, fast-filtering BI models.

Table	Role	Key Columns
Fact_Crimes	Fact (grain: State–District–Year–Category)	STATE_UT, DISTRICT, YEAR, CrimeCategory, CrimeGroups, Cases
Dim_Year	Dimension	Year (2001–2012)
Dim_State	Dimension	STATE_UT (35 distinct states)

Step 5 — Write and debug DAX measures

Around 20 measures were built for totals, growth rates, and rankings. Several needed a second pass after producing visibly wrong numbers in testing — documented in full in Section 5.

3 Core Metrics & DAX

Headline KPIs (validated against pandas, not just DAX output)

23.9M	2.65%	2.76%	8.33%
TOTAL CRIMES	YOY CHANGE	CAGR 2001–12	VIOLENT CRIME %

Metric	Value	Verified via
Total Crimes (2001–2012)	23,940,654	pandas sum, cross-checked against TOTAL_IPC_CRIMES column
CAGR (2001–2012)	2.76%	pandas (start/end year totals) and DAX, matched exactly
YoY Change (2011 → 2012)	2.65%	pandas year-over-year calc, matched DAX measure
Violent Crime %	8.33%	pandas category-group sum ÷ total
Top State by Volume	Madhya Pradesh — 2,413,770 cases	pandas groupby, matched DAX ranking measure

Selected DAX Measures

```
Total Crimes = SUM(Fact_Crimes[Cases])

CAGR 2001-2012 =
VAR StartVal = CALCULATE([Total Crimes], Dim_Year[Year] = 2001)
VAR EndVal   = CALCULATE([Total Crimes], Dim_Year[Year] = 2012)
RETURN (EndVal / StartVal) ^ (1/11) - 1

Violent Crime % =
DIVIDE(
    CALCULATE([Total Crimes], Fact_Crimes[CrimeGroups] = "Violent Crime"),
    [Total Crimes]
)
```

```
State Rank =
VAR CurrentState = SELECTEDVALUE(Dim_State[STATE_UT])
RETURN IF(
    HASONEVALUE(Dim_State[STATE_UT]),
    COUNTROWS(FILTER(ALL(Dim_State[STATE_UT]),
        CALCULATE([Total Crimes]) > CALCULATE([Total Crimes],
            Dim_State[STATE_UT] = CurrentState))) + 1,
    1
)
```

Ranking measures use a manual COUNTROWS/FILTER pattern instead of RANKX, after RANKX produced incorrect results when it evaluated over the unpivoted fact table's row-level grain rather than the distinct-district grain (see Section 5).

4 Dashboard Design

The final deliverable is a 3-page interactive Power BI dashboard, built on a consistent design system rather than default visuals.



Page 1 — Crime Analysis (National Overview)

- KPI row: Total Crimes, YoY Change, CAGR, Violent Crime %, Top State
- National trend line (2001–2012), crime-category breakdown (Top 10), state-level heat matrix
- Left control panel: Date Range, State, District, Crime Category slicers

Page 2 — State Analysis

- KPI row: Total Crimes (state), State Rank, Districts Covered, Top District
- Top 10 States and Top 10 Districts bar charts, state trend line, State × CrimeGroup cross-tab matrix with a conditional-formatting color scale

Page 3 — District Analysis

- KPI row: District Total, District Rank (within state), Crimes per District (state average), Violent Crime %
- District trend line, Top 10 Districts funnel chart, CrimeGroups donut chart, District-vs-State-Average gauge
- Cross-page navigation buttons linking all three pages

Design System

Element	Spec
Color palette	Navy #2E4057 (primary) · Red #C0392B (alert/high) · Amber #F1C40F (neutral) · Green #27AE60 (good)
Typography	Segoe UI / Segoe UI Semibold throughout
KPI cards	White background, rounded corners, soft shadow, left color-accent bar
Layout grid	Header · KPI row · Main visuals · Footer, consistent across all 3 pages

5 Debugging Log

Kept intentionally — the debugging process is as much a part of the analytical skill demonstrated here as the final numbers.

Issue	Root Cause	Fix
Total Crimes inflated to 28M+	Sub-category columns double-counted with parent category	Excluded 6 overlapping sub-columns in Power Query
YoY % showing 929.5%	Measure evaluated without a Year filter context (grand total vs. single prior year)	Added explicit Latest-Year / Prior-Year measures independent of visual context
CAGR displaying as 0	Correct value (2.76%) rounded to 0 under General Number format	Set measure format to Percentage
State/District Rank always 1	RANKX evaluated with no single-value filter context (slicer on "All")	Added HASONEVALUE guard + manual COUNTROWS ranking pattern
District Rank showing 190,864	RANKX ranked raw unpivoted fact rows instead of distinct districts	Ranked over CALCULATETABLE(VALUES(DISTRICT), ...) instead of the full fact table
Gauge target showing no gap	State-average measure was still filtered by the selected district	Wrapped in CALCULATE(..., ALL(DISTRICT)) so it always reflects the full state
Map / Filled Map visuals disabled	Org-level Power BI tenant policy blocking Bing-Maps-based visuals	Replaced with a conditionally-formatted Matrix as a map alternative

> **Why this matters:** every one of these was caught by comparing DAX output against an independent pandas calculation, not by assuming the first number was correct. That habit is the actual skill being demonstrated — not any single chart.

6 Key Insights

- Crimes grew at a **2.76% CAGR** from 2001–2012, with a visible acceleration after 2007.
- **Theft-related offenses** (Theft + Burglary) form the largest identifiable property-crime block nationally.
- **Madhya Pradesh** recorded the highest total crime volume over the period (2,413,770 cases), narrowly ahead of Maharashtra (2,273,436).
- Violent crime accounts for a relatively small share (**8.33%**) of total reported IPC crime — the bulk falls under property and "other" IPC categories.
- District-level analysis shows substantial variance even within the same state — several districts sit well above their state's per-district average, useful for targeted resource allocation.

> **Headline finding:** Property crime is the single largest, fastest-growing, and most consistently reported crime category in the dataset — making it the highest-leverage area for policy or resource-allocation focus, ahead of violent crime which draws more public attention but represents a much smaller share of the total.

7 Skills Demonstrated

Category	Skills
Data Engineering	SQL (MySQL), Power Query (M), data cleaning, deduplication, schema design
Analysis	Python (pandas) — profiling, aggregation, validation, root-cause analysis of data-quality issues
Modeling	Star schema design, DAX (context transition, CALCULATE/FILTER patterns, ranking logic)
Visualization	Power BI dashboard design, KPI structuring, conditional formatting, multi-page navigation
Analytical Thinking	Reconciling conflicting totals, root-causing DAX context bugs, deriving a defensible headline insight from raw data